

HUISEOK MOON

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Research Interests

My research combines **Assistive Robotics** and **Physical Human-Robot Interaction**, with the foundation of my experience in the design of clinically validated real-time intent detection frameworks and assistive robotic control based on sensor systems (EMG, IMU, etc.). I am interested in physics-based modeling and simulation, model-based control, and multimodal AI for understanding and predicting human motion, with a focus on mobility assistance for wearable robotics.

Keywords: Wearable Robotics, Assistive Robotics, Multimodal Learning, Nonlinear Control

Education

Nov 2019 – Jan 2024	Ph.D. in Signal, Image and Control — Université Paris-Est Créteil, France Advisor: Dr. Samer Mohammed Thesis: <i>Human-in-the-loop Paradigms for Mobility Assistance Using Powered Exoskeleton</i>
Mar 2017 – Feb 2019	M.S. in Mechanical Engineering — Sogang University, Seoul, South Korea Advisor: Dr. Kyoungchul Kong Thesis: <i>Control Algorithm of a Robotic Leg for Reducing Interaction Force Between a Load and the Human Body</i> Graduate research fellowship, Ilwun Science and Technology Foundation (2017–2019)
Mar 2010 – Feb 2017	B.S. in Robotics (Intelligence System) — Kwangwoon University, Seoul, South Korea Thesis: <i>Research on personal mobility using ball balancing robot</i>

Professional Experience

Jul 2026 – Present	Postdoctoral Researcher — Université Paris-Est Créteil, France - Designed a high-precision gait mode recognition system using LSTM networks, achieving 98% classification accuracy and enabling real-time intent detection for actuated ankle-foot orthoses. - Developed an energy-efficient adaptive FES control strategy based on surface EMG signals, reducing stimulation energy consumption by 30% while mitigating muscle fatigue in clinical trials. - Implemented a multi-sensor fusion framework (7 IMUs) for lower-limb joint analysis and Parkinson's disease diagnosis in a portable clinical device.
Sep 2025 – Mar 2026	Postdoctoral Researcher — KAIST, Daejeon, South Korea (Advisor: Dr. Jaegul Choo) Developed physics-informed AI models for human motion intention inference, leveraging motion-primitive representations of wearable IMU signals.
Sep 2023 – Aug 2025	Research Engineer — Université Paris-Est Créteil, France - Gait mode recognition (LSTM, 98% accuracy) for real-time intent detection on actuated ankle-foot orthoses. - Energy-efficient adaptive FES control validated in clinical trials. - Multi-sensor fusion framework (7 IMUs) for clinical gait analysis and Parkinson's disease diagnosis.
Nov 2022 – Aug 2023	ATER (Teaching & Research Assistant) — Université Paris-Est Créteil, France Undergraduate lectures and hands-on labs on wearable sensor integration (EMG, IMU, FSR) and biomechanical signal processing using Python and MATLAB.
Nov 2019 – Oct 2022	Research Assistant — Université Paris-Est Créteil, France

- Nonlinear control architectures (impedance control, reference tracking) for human-exoskeleton interaction, applied to lower-limb exoskeletons and sit-to-stand assistance robots.
- Real-time activity recognition pipeline combining Random Forests and deep learning with Kalman-filtered IMU features.

Mar 2017 – Feb 2019

Research Assistant — Sogang University, Seoul, South Korea

- Interaction force control strategies for lower-limb exoskeletons for load-carrying tasks.

Publications

Journal articles

- 2025** Real-Time LSTM-Driven Dynamic Gait Mode Detection for Enhanced Control of Actuated Ankle-Foot Orthosis.
H. Moon, O. Bey, A. Boubezoul, L. Oukhellou, S. Mohammed. *IEEE Transactions on Robotics*, 41, 4794–4809.
- 2023** Fuzzy Convolutional Attention-based GRU Network for Human Activity Recognition.
G. Khodabandelou, **H. Moon**, Y. Amirat, S. Mohammed. *Engineering Applications of Artificial Intelligence*, 118, 105702.
- 2022** Hybrid Half-Gaussian Selectively Adaptive Fuzzy Control of an Actuated Ankle-Foot Orthosis.
H. Moon, R. Maiti, K. Das Sharma, Y. Amirat, P. Siarry, S. Mohammed. *IEEE Robotics and Automation Letters*, 7, 9635–9642. (ICRA 2023 oral)
- 2022** An Assistive Explicit Model Predictive Control Framework for a Knee Rehabilitation Exoskeleton.
I. Jammeli, A. Chemori, **H. Moon**, S. Elloumi, S. Mohammed. *IEEE/ASME Transactions on Mechatronics*, 27, 3636–3647.
- 2022** Impedance Modulation Control of a Lower-Limb Exoskeleton to Assist Sit-to-Stand Movements.
W. Huo, **H. Moon**, M. A. Alouane, V. Bonnet, J. Huang, Y. Amirat, R. Vaidyanathan, S. Mohammed. *IEEE Transactions on Robotics*, 38, 1230–1249.

Conference proceedings

- 2024** A Novel Funnel-Based L1 Adaptive Fuzzy Approach for the Control of an Actuated Ankle Foot Orthosis.
O. Bey, R. Jradi, **H. Moon**, H. Rifai, K. Das Sharma, Y. Amirat, S. Mohammed. *IEEE ICRA*.
- 2022** Online Human Intention Detection through Machine-learning based Algorithm for the Control of Lower-limbs Wearable Robot.
H. Moon, A. Boubezoul, L. Oukhellou, Y. Amirat, S. Mohammed. *IEEE-RAS Humanoids*.
- 2021** A Novel Gait Phase Detection Algorithm for Foot Drop Correction through Optimal Hybrid FES-Orthosis Assistance.
P.-G. Jung, W. Huo, **H. Moon**, Y. Amirat, S. Mohammed. *IEEE ICRA*.

Academic Service & Teaching

- Peer review** IEEE Transactions on Robotics, IROS, ICRA, Engineering Applications of Artificial Intelligence, Mechatronics, etc.
- Invited talks** 2025: REHAssist Visiting Seminar, EPFL (Switzerland) · 2024: LISSI-BIOTN Joint Seminar, UPEC (France) · 2020–2023: LISSI Research Seminars, UPEC (4 sessions).
- Volunteering** 2025: IFAC World Congress · 2024: IUT Research Day · 2023: Fête de la Science · 2017: Korea Defense Science & Tech Fair.

Teaching	May 2022 – Aug 2023: ATER, UPEC — core Python and multimodal sensor integration (EMG, IMU, FSR).
Mentoring (M.S.)	L. Marusu (2025), A. A. Benhamlaoui (2024), D. Plet (2024), W. Legendre (2023), J. M. Fadous (2022), R. Ravaka & M. Diaby (2021).

Skills

Programming	Python, C, C++, MATLAB, LabVIEW, Solidworks
Frameworks & tools	PyTorch, TensorFlow, Simulink
Expertise	Real-time ML deployment, multimodal sensor fusion, embedded systems, adaptive & impedance control, human-robot interaction, wearable robotics, signal processing
Languages	Korean (native), English (fluent), French (basic)

References

Postdoc advisor	Jaegul Choo — Associate Professor, Kim Jaechul Graduate School of AI, KAIST · jchoo@kaist.ac.kr
Ph.D. advisor	Samer Mohammed — Associate Professor, LISSI, Université Paris-Est Créteil · samer.mohammed@u-pec.fr
M.S. advisor	Kyungchul Kong — Associate Professor, KAIST · kckong@kaist.ac.kr